



# Effects of an adaptive computer agent-based digital game on EFL students' English learning outcomes

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## Abstract

The effectiveness of digital game-based language learning (DGBLL) has been recognized by scholars. With the development of computer technology and multimedia learning environments, computer agents have been widely used in game systems to provide learning guidance or assistance. A computer agent is a virtual character created in a digital learning system to achieve instructional goals. However, in traditional teaching systems, computer agents are designed to perform a single role, such as a teacher or a student. Computer agents with a single interactive logic often lack the functionality required to understand students' conditions and needs from various perspectives, and cannot adapt for better learning support. Therefore, this study proposed an adaptive role-switching strategy that allows computer agents to adjust their roles and functions according to students' needs in digital games to promote their learning achievement. An adaptive computer agent-based digital game was developed to investigate the impact of this model on English vocabulary learning achievement, motivation, self-efficacy, and English anxiety among EFL (English as Foreign Language) students. A quasi-experiment was designed and 56 middle school students in four classes were recruited. Two classes ( $n=30$ ) were arranged to be the experimental group which used an adaptive computer agent-based digital game (adaptive computer agent-based digital game, ACA-DG), while two classes ( $n=26$ ) were arranged to be the control group which used the conventional computer agent-based digital game (conventional computer agent-based digital game, CCA-DG). The results showed that students in the experimental group had significantly higher learning achievement and self-efficacy than those in the control group. On the other hand, students in the experimental group had significantly lower English anxiety than those in the control group. There was no significant difference between the two groups regarding learning motivation.

**Keywords** Computer agent · Adaptive learning · Digital game-based learning · Language learning

## Introduction

When learners encounter learning difficulties or make mistakes in language learning, negative emotions, such as depression and frustration, may lead to learning anxiety (Wei et al., 2018; Zou et al., 2021), which could interfere with language input and adversely affect their learning engagement and self-efficacy (Reinders & Wattana, 2015; Yang et al., 2018). With the development of computer technology, digital game-based language learning has become a common area of research and is considered to complement the shortcomings in conventional language learning (Chen & Hsu, 2020; Zou et al., 2021). Extensive game scenarios can facilitate students' scientific exploratory behaviors, such as searching, observing, and interacting, and encourage students' knowledge construction and learning performance (Wei et al., 2018; Zeng et al., 2020). Well-designed interfaces, tasks, and background stories positively affect learners and increase their interest and engagement in learning (Mostefai et al., 2019). Students performed better on content knowledge assessment and had higher self-efficacy in a game-based learning approach than in conventional lecture-based instruction (Wang & Zheng, 2021).

However, digital games also have several challenges. For example, during the gaming process, players' cognitive load and learning anxiety could be increased when trying to deal with the amount of auditory and visual information (Davis et al., 2021). Scholars have indicated that too many task objectives are likely to distract students and reduce their learning efficiency during the game-based learning process (Shute et al., 2021). In addition, low-interest digital games only sustain short-term learning effects, and learners may not be as motivated as expected (Tanaka, 2017). Therefore, not every learner benefits from a game-based learning environment (Yang et al., 2018). Teachers should consider inducing positive emotions in the game design to maximize the benefits of the game-based learning experience for students.

Adaptive learning has become a potential solution to improve the outcomes of digital learning systems (Pai et al., 2021). Adaptive learning is a way to provide personalized instruction or feedback based on the needs of learners (El-Sabagh, 2021). In conventional digital educational games, teachers only offer the same material to all students, which makes it challenging to consider the needs of different students (Hwang et al., 2020). From a developmental perspective, creating better adaptive computer agents based on students' play behaviors and cognitive profiles benefits students' skill acquisition and emotional engagement. Hwang et al. (2020) developed a fuzzy expert system-based adaptive learning system to help low-achieving students reduce their learning anxiety and complete the learning tasks successfully. Chen et al. (2020) used a role-transfer model to personalize agent strategies, and found that two-way reciprocal agent peers maximized the impact on children's vocabulary learning and emotional engagement.

So far, most computer agents in teaching systems have been designed to perform a single role, such as teacher, peer, or student. However, this design lacks the flexibility to make personalized adjustments based on students' conditions and to serve as a teaching tool. Therefore, for this study we developed an adaptive computer agent-based digital game. The computer agent will dynamically switch between various styles according to the student's behavior and experience in the game. This study aimed to investigate the effects of adaptive computer agents on EFL students' learning performance, cognition, and behavior in the game, and to provide a reference for future educational game design. The research questions posed in this study are as follows:

- (1) Do EFL students who learn with the adaptive computer agent-based digital game (ACA-DG) have higher English learning achievement than those who learn with the conventional computer agent-based digital game (CCA-DG)?
- (2) Do EFL students who learn with the ACA-DG have higher learning motivation than those who learn with the CCA-DG?
- (3) Do EFL students who learn with the ACA-DG have higher self-efficacy than those who learn with the CCA-DG?
- (4) Do EFL students who learn with the ACA-DG have lower English anxiety than those who learn with the CCA-DG?
- (5) Do EFL students studying with the CCA-DG and the ACA-DG have different behavioral features?

## Literature review

### Game-based language learning

Game-based learning is an approach incorporating game elements (e.g., competition, interaction) and rules to achieve quantifiable outcomes (e.g., wins or scores) and educational goals (Hung et al., 2018; Shute et al., 2021). It has a variety of manifestations, including the use of narratives to change the traditional framework of the task, the creation of social competition, and the encouragement of desired behaviors through gaming missions and contexts (Hanus & Fox, 2015). A digital game learning environment has three features: simulation, interactivity, and sense of presence (Zeng et al., 2020). Simulation refers to the reproduction of real situations across spatial barriers which build bridges between virtual scenes and real life (Qian & Clark, 2016). Interactivity increases students' social interaction in games and reduces learning anxiety (Chen & Hsu, 2020). The sense of presence permits students to immerse themselves in the game environment to observe, explore, and experience (Reinhardt, 2017). Therefore, game-based learning has been widely used in the classroom to increase learners' interest and clarify their educational goals (Bunyakul et al., 2022; Daungcharone et al., 2020; Komalawardhana et al., 2021; Yadav & Oyelere, 2021).

In addition, game-based language learning benefits students' knowledge integration and higher-level thinking development (Qian & Clark, 2016). Chen et al. (2020) investigated the effect on second language vocabulary acquisition by using L1 keywords as subtitles in an adventure game. The results showed that additional L1 vocabulary support improved students' attention and learning performance. As learning motivation plays a crucial role in enhancing language acquisition and achievement, Tsai et al. (2017) explored the predictability of the impact of learners' achievement on motivation. The results showed that game-based learning significantly increased students' motivation and provided external stimulation to improve their learning efficiency. Enriched multimedia materials and interactive methods also increase learners' engagement and willingness to communicate in the target language (Reinders & Wattana, 2015; Reinhardt, 2017). Chats and interactions increase the opportunities for language output as well as help to increase students' intrinsic motivation and reduce their target language anxiety (Yang et al., 2018).

Although many researchers have supported the positive effects of play-based learning, educators still face many challenges (Hung et al., 2018). Students become frustrated when they consistently fail at complex challenges or tasks (Rahimi et al., 2021). Games should have complete motivational rules and strategies to guide students in the behaviors that

educators expect (Yang et al., 2016). Many games that promote subject education provide relevant curriculum information and text, but lack engagement and fun (Yadav & Oyelere, 2021). Conversely, excessively entertaining designs negatively affect learning outcomes (Kim et al., 2009).

Educators incorporate learning strategies into game-based learning to balance playfulness and instruction. For example, Israel-Fishelson and HersHKovitz (2021) evaluated students' problem solutions and gave them different numbers of stars to motivate them to continuously improve their solutions and obtain more rewards. Yang et al. (2016) designed a badge system to overcome students' lack of self-efficacy. This system included digital badges, leaderboards, and multimedia icons as game elements. The results showed that the badge system significantly positively affected learners' self-efficacy and English learning performance. It supports the idea that incorporating learning strategies into game-based learning is an effective way to enhance teaching effectiveness.

### Computer agents in game-based learning

Owing to the rapid advancement of computing and multimedia technologies, using computer agents to mimic humans to facilitate students' digital learning has become a potential approach to improving students' learning engagement and efficiency (Schroeder & Gotch, 2015). Computer agents are defined as computerized human-like characters in multimedia environments that drive behaviors using rules and technology (Dinçer & Doğanay, 2017; Richards & Dignum, 2019). As a learning-supported technology, computer agents have been designed in diverse learning systems, such as intelligent tutoring and gaming applications, to support students of different ages or application domains (Belpaeme et al., 2018; Schroeder et al., 2017). They typically play the role of a learning assistant to provide help and guidance to learners or to demonstrate complex principles or processes (Martha & Santos, 2019).

Social agency theory can be used as a theoretical framework for the effectiveness of computer agents in modeling human interactions and connections. This theory suggests that verbal and visual cues in a computer-based environment can meet learners' expectations for interpersonal interaction and encourage them to establish social partnerships (Atkinson et al., 2005). In addition, Maslow's (1943) motivation theory also emphasizes that, in social learning contexts, in-depth interactions between individuals lead to better outcomes. Thus, in learning contexts, interactive computer agents promote students' willingness to communicate and study (Castro-Alonso et al., 2021). Computer agents establish social partnerships with learners through voice or image, leading to social reflection, and improve students' learning behaviors (Davis et al., 2021). For example, students who interact with competitive computer agents show a greater desire to share, and the ability to use logic to solve new problems (Okita & Schwartz, 2013). Sawyer et al. (2017) found that students with high agency in game-based learning had higher engagement and better learning outcomes.

In addition, the effectiveness of the teaching system is affected by the interaction logic and style of the computer agents with the learners. Common agent styles include peer agents (same skill level as the player), student agents (less skilled), and teacher agents (more capable) (Cai et al., 2014; Chen et al., 2020). For instance, the teacher computer agents observe students' learning process and provide real-time help (Ma et al., 2014). Students are more likely to obtain knowledge and skills from a knowledgeable agent as a teacher (Belpaeme et al., 2018). The computer agent as a peer can provide two-way

reciprocal teaching strategies (Howard et al., 2017). The peer agents guide learners to think deeply about the connections between data and the meaning behind them (Schroeder et al., 2017). On the other hand, the computer agent as a student has more group affinity attributes that can stimulate learners' interest and increase their knowledge retention (Tanaka, 2017). It improves learners' knowledge and higher-level thinking through feedback, and encourages them to act as teachers or advisors (Tanaka, 2017). The agent's linguistic style also impacts learning. According to Bandura's (2012) social learning theory, the phenomenon of learning is a process of imitation and substitution based on observing others' behaviors and outcomes. The computer-mediated interactions amplified the correlation between disclosure and intimacy in contrast to face-to-face interactions (Jiang et al., 2011). This enhancement phenomenon was entirely mediated by the augmented interpersonal attributions observed during the computer-mediated condition. Thus, exploring the interaction patterns of computer agents can help students create emotionally engaging experiences and maximize the outcomes of instructional systems (Chen et al., 2020).

### Adaptive learning in game-based learning

Providing personalized learning support to improve students' learning achievement is an essential issue in the educational application of computer technology (Hwang et al., 2020). Adaptive learning enhances the learning experience by making adjustments based on learners' behavioral performance or evaluation results (El-Sabagh, 2021). Arguedas and Daradoumis (2021) suggested that adaptive learning adjusts learning progress and content according to learners' abilities, interests, emotions, and other characteristics during the learning process. A student-centered adaptive learning model considers students' differences and achieves better learning outcomes (Schrader & Grassinger, 2021). Bevacqua et al. (2011) provided different types of feedback based on the user's verbal and nonverbal behavior in response to the learner's emotional and cognitive state. Liu and Graf (2009) proposed an adaptive learning approach that provides support and cues to students who need help based on their learning style and behavior. Thus, most adaptive learning systems use cognitive or affective states as indicators of model adjustment (Richards & Dignum, 2019).

Many researchers have attempted to develop adaptive learning systems based on affective expressions. Hwang et al. (2020) used an expert system to analyze students' affective states and knowledge levels to determine individual learning goals. Karampiperis and Sampson (2005) developed multiple learning paths in an instructional system and then selected the most appropriate path based on the students' affective and cognitive levels. Besides, many studies support that adaptive digital learning environments based on students' affective and behavioral factors contribute to students' motivation and engagement (El-Sabagh, 2021; Schrader & Grassinger, 2021; Standen et al., 2020).

Technological advances allow agents to detect users' emotional states and customize interaction styles using external or internal features (Taub et al., 2020). Computer agents are used in learning systems to imitate human tutors to answer student questions or provide feedback (Dinçer & Doğanay, 2017). Graesser et al. (2017) attributed agent effectiveness to a systematic conversational structure called expectation and misconception-tailored (EMT) dialogue. When an agent poses challenging questions to a student, it expects specific correct and incorrect answers. The agent forms a model of the student's knowledge acquisition based on the answers presented in multiple conversational rounds. Subsequent

conversations are generated from the model, such as encouraging students to produce content or correcting misconceptions.

Previous educational games tended to ignore emotional regulation, so investigating the impact of different styles of agents on students' affect has become an essential consideration in designing educational games (Pacella & López-Pérez, 2018). Johnson and Lester (2018) found that agents who demonstrated empathy significantly increased learners' perceptions of presence in a virtual environment. Chen et al. (2020) developed a bi-directional reciprocal learning model that allowed students to experience teacher agents and student agents to strengthen emotional ties. Emotional feedback from the agent boosts students' curiosity, creativity, and confidence in completing learning tasks while reducing negative emotions such as boredom and anger (Arguedas & Daradoumis, 2021). In conclusion, developing an agent-switching strategy that adapts to individual needs significantly benefits student performance and cognition. Therefore, this study developed an agent-switching strategy based on social agency theory and social cognitive theory. We hoped to improve students' learning performance, motivation, and self-efficacy through dynamic adjustment of computer agents in digital game-based language learning.

## **An adaptive computer agent-based digital game system**

### **The adaptive computer agent mode**

This study attempted to develop an adaptive computer agent-based digital game (ACA-DG) to help EFL students improve their English literacy skills. The computer agent in the game guides students to complete the tasks and to learn about the language. The computer agent in the game has three roles: teacher, peer, and student. Each of the three roles interacts with the player in different modes, as shown in Table 1. The adaptive nature is reflected by the switching of agent roles when players are learning in the game. In the teacher role, the computer agents have a high level of awareness. Many scholars have suggested adding learning support as scaffolding to help learners improve their game self-efficacy (Rahimi et al., 2021; Sun et al., 2018). The teacher agents master all the task details and have sufficient experience and confidence. When players fail repeatedly or become frustrated, the teacher agents provide real-time content supports, such as guidance or suggestions. Players have less flexibility with teacher agents than with other styles of computer agents.

In the peer role, the computer agent is presented as a peer. When players encounter problems, peer agents give 80% of the correct information and do not show prominent emotional characteristics. Peer agents give players more opportunities to interact than teachers and student agents. Depending on the situation, players can choose whether or not to seek help. Also, peer agents give suggestions for review when tasks fail.

In the student role, computer agents show immaturity or lack of confidence. Previous research showed that for higher-level learners, acting as a teacher increased their interest in learning (Sawyer et al., 2017; Tanaka, 2017). Therefore, the student agents repeatedly confirm the players' mastery of the task by asking questions. When players perform well, the student agents do not interfere too much. When players lose the target or encounter difficulties, 50% of the clues provided by the student agents are correct. If a task repeatedly fails, the student agents will provide suggestions for review. Players receive more encouragement and interaction from the student agents.

**Table 1** Computer agent interaction mode

| Category  | Teacher            | Peer                                                                                                   | Student                                                                                                                           |
|-----------|--------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Cognition | Task description   | Explain the details of the task requirements                                                           | Express a misunderstanding of the key words in the task requirements                                                              |
|           | Tips               | Always give players the right tips                                                                     | Give players tips that are 50% correct                                                                                            |
|           | Target collection  | Express sufficient confidence in collecting targets<br>(e.g., "I know which target is the right one.") | Expressing lack of confidence when collecting targets<br>(e.g., "I'm not sure which target is the right one.")                    |
|           | Answer hints       | Provide hints when the player gives incorrect answers                                                  | Without hints                                                                                                                     |
| Emotion   | Review suggestions | Before the task begins<br>(e.g., "Are you sure you don't need a review?")                              | Task failed with more than 5 wrong answers<br>(e.g., "I think the most important thing to do is to review the knowledge points.") |
|           | Correct answer     | Pleased/proud<br>(e.g., "You got it right, congratulations!"<br>"You deserve to be my student")        | Admiration/excitement<br>(e.g., "I can't believe we did that."<br>"Maybe you should teach me how you did it.")                    |
|           | Incorrect answer   | Rectify<br>(e.g., "Let's correct it together.")                                                        | Encouragement/support<br>(e.g., "We all make mistakes." "You'll be right next time. I believe in you.")                           |
|           |                    |                                                                                                        |                                                                                                                                   |

## Adaptive role-switching strategy

An expert decision-oriented adaptive role-switching strategy was developed to determine the style of the computer agent based on the players' behavioral information in the game. To construct a scheme for adaptive role switching, we built a decision tree based on the judgment of 30 cases provided by two English teachers with many years of experience. As shown in Table 2, the feature codes used in this study were constructed concerning the affective and cognitive models from the studies of Hwang et al. (2020) and Chen et al. (2020). Cognitive status refers to the player's learning achievement when completing the task. Help-seeking refers to players seeking advice or support from the computer agent after encountering problems. Non-task interaction refers to the interaction between players and non-target NPCs or exploring the environment during the game. Knowledge review refers to whether the player chooses to review knowledge during the task process.

The decision tree model is shown in Fig. 1 after combining the opinions of the two experts and testing the validity. The computer agent dynamically adjusts the interaction style and logic during the game according to the behavior features of the players. There are six branches in the decision tree, which correspond to six situations. (1) Players with fewer errors and fewer support requests have good independent learning performance and a higher cognitive level. They are capable of assisting students with lower abilities. Therefore, the computer agent model changed to the student style. (2) Players with fewer errors, more support requests, and more non-task interactions have good learning performance. They show a positive tendency to explore and interact. Therefore, the computer agent model changed to the student style. (3) Players with fewer errors, more support requests, and fewer non-task interactions have good learning performance but need to strengthen their social behaviors. Therefore, the computer agent model changed to the peer style to encourage them to interact with the agent and the environment. (4) Some players with more errors and more support requests choose to do knowledge reviews. Although their learning performance is not as expected, they tend to pursue knowledge actively. Therefore, the computer agent changes to the peer style to facilitate the players' adaptive adjustment. (5) Some players with more errors and more support requests choose to skip the knowledge reviews. Their learning performance needs to be improved, and they failed to actively pursue knowledge. Therefore, the computer agent changes the teacher style to increase hints and suggestions. (6) The performance of players with more errors and fewer

**Table 2** Player behavior feature codes

| Category             | Code  | Description                                                                                        |
|----------------------|-------|----------------------------------------------------------------------------------------------------|
| Cognitive status     | CS-H  | Fewer errors than the average of the pretest students                                              |
|                      | CS-L  | More errors than the average of the pretest students                                               |
| Help-seeking         | SH-H  | More support requests than the average of the pretest students                                     |
|                      | SH-L  | Fewer support requests than the average of the pretest students                                    |
| Non-task interaction | NTI-H | More conversations or interactions with non-task targets than the average of the pretest students  |
|                      | NTI-L | Fewer conversations or interactions with non-task targets than the average of the pretest students |
| Knowledge review     | RK-Y  | Select to review                                                                                   |
|                      | RK-N  | Select to skip the review                                                                          |

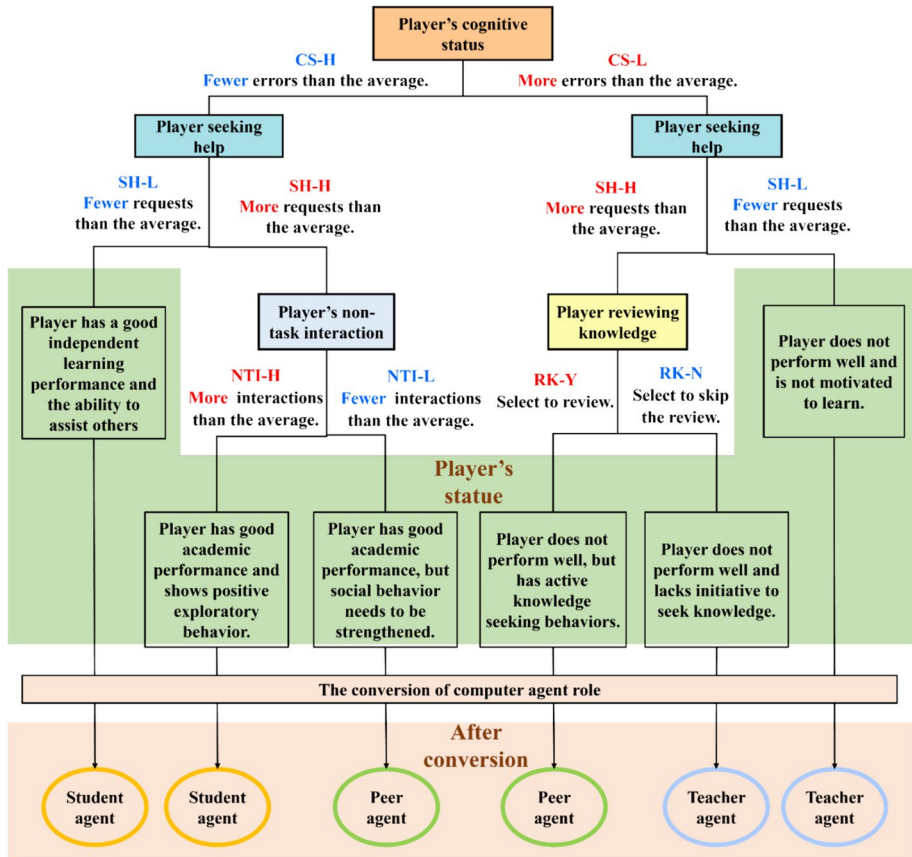
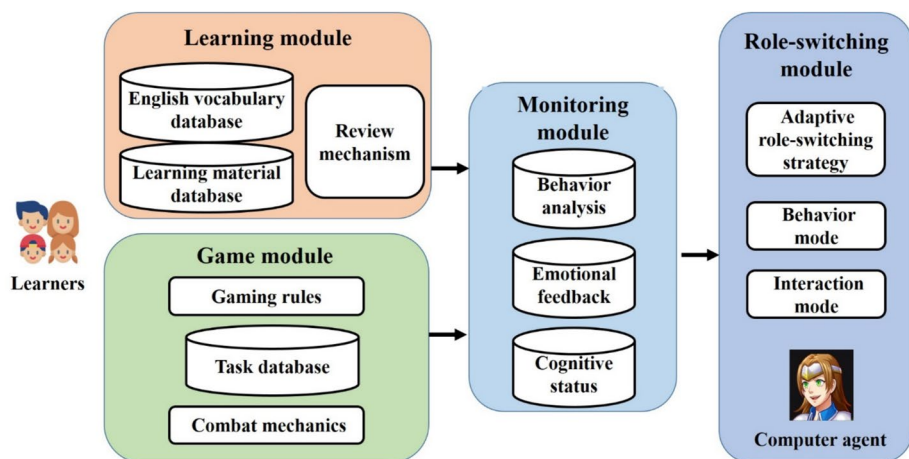


Fig. 1 The decision tree model

support requests needs to be improved. They treat learning negatively. Therefore, the computer agent converts to the teacher style to provide learning support and advice and regulate the players' behavior.

## System development

The game system in this study was produced using the RPG Maker MV developed by Enterbrain Incorporation and incorporated game elements such as challenges, battles, and treasure hunts. Figure 2 illustrates the system structure of the adaptive computer agent-based digital game (ACA-DG). Players experienced a learning module, a game module, a monitoring module, and a role-switching module in the game. In the learning module, the researcher developed an English vocabulary database referring to a section about prepositional words in a junior high school English textbook. Non-player characters (NPCs) in the game teach students about English vocabulary through dialogues. The learning materials database provides instructional pictures or videos to help players understand. Once the players have finished the lesson, they can use a review mechanism to return to the lesson at any time.



**Fig. 2** The system structure of ACA-DG

The game module informs players about the game's operation and rules. The system collects and analyzes the students' behavioral, emotional, and cognitive status in the monitoring module. Behavioral information includes learned behaviors (e.g., conversations with target NPCs, review behaviors) and non-learned behaviors (e.g., environmental exploration, useless interactions). Emotional feedback is students' concentration and perseverance by recording their help-seeking or trial-and-error behaviors while answering questions. The system also measures students' cognitive status of English vocabulary through their accuracy in answering the questions.

In the role-switching module, the system uses the decision tree model to analyze the collected information and decide whether to switch agent styles. The computer agent acts as a partner to the players throughout the game and works with them to complete tasks, providing help and guidance at the appropriate time. The system has three styles of computer agents: teacher, peer, and student. In the experimental group, the system presets the style of the computer agent to the peer style after the game starts. At the end of each task, the role-switching module analyzes the players' current behavioral and cognitive status. Based on the results, players with insufficient learning in the round are adjusted to the appropriate computer agent mode. The computer agents in the game switch behaviors and interaction patterns to match one of the styles. Role-switching aims to maximize the teaching effectiveness of game-based learning through computer agents that balance freedom and learning support in the game.

The conventional computer agent-based digital game (CCA-DG) used in the control group does not have a monitoring module or a role-switching module. Since most agents in traditional teaching systems are cognitive tutors (Pai et al., 2021), the default computer agent in the control group was a teacher agent to fit the general educational game state. Compared with the experimental group, the computer agent used in the control group does not have the capability of adaptive role-switching.

Once the game starts, the computer agent first introduces the game's backstory, rules, and ultimate goal to the players. Players are also informed of the functions of the computer agent as a partner. After players enter the tutorial stage and learn prior knowledge, they need to complete three different tasks assigned by the NPCs. The game ends after all



Fig. 3 The interface of the tutorial stage

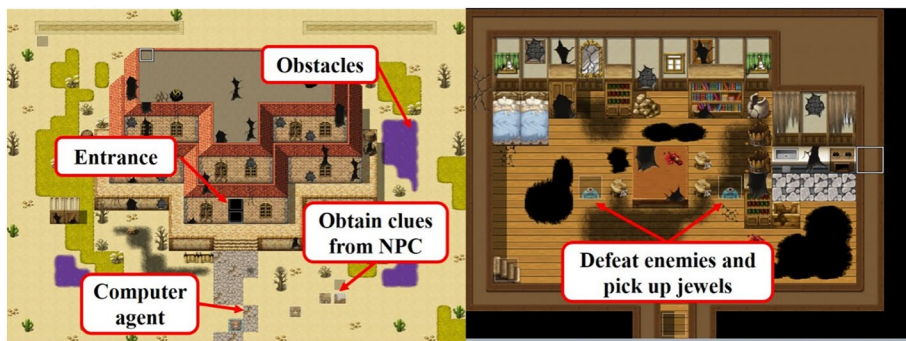


Fig. 4 The interface of the "treasure hunt" task

tasks are completed. Between each task round, the game system used in the experimental group determined if the computer agent's role needed to be changed based on the player's behavioral, emotional, and cognitive performance. The computer agent in the control group maintained the teacher role throughout the game.

As shown in Fig. 3, the NPCs teach the players about the prepositions that appear in the textbook through dialogues. The players learn two types of preposition: time prepositions (fire magic) and location prepositions (water magic). When the NPCs explain the knowledge, they display pictures or videos with the learning database to help players understand. After the players have learned all the content, they can review it again at any time.

Figure 4 shows the game interface of the treasure hunt. The players need to overcome the obstacles and find the target NPC with the help of the computer agent who prompts the players with the target vocabulary that appeared in the previous tutorial stage. Players must use their prior knowledge to analyze the hints and find the correct route. It is designed to test whether the player can correctly understand the meaning of the target term. After correctly answering the questions posed by the system, the player can defeat the enemies and collect the dropped jewels.

The food collection task is shown in Fig. 5. The players need to interact with the target item and trigger a set of multiple-choice questions. If the players answer the



Fig. 5 The interface of the “food collection” task

questions correctly, they can collect the food; if the answer is wrong, the food collection will fail. The task requirement is to collect five food items successfully. The mission will fail if the cumulative number of incorrect answers reaches five. Players must go back to the tutorial stage to review and restart the collection task.

The last task is to defeat the enemies, as shown in Fig. 6. The player has to enter the cave to engage with various monsters and trigger the battle interface. Different types of magic correspond to target words, including time and place prepositions. The attack system consists of multiple-choice questions, each with only one correct answer. Players can perform the corresponding magic to attack or defend when they select the right option. Enemies will alternately attack and deduct time from the player’s life. If the players defeat all enemies, the mission is successful; if their life is reduced to 0, they fail the task. The computer agent will follow the player as a partner during all tasks. The player can interact with the computer agent in different ways, such as asking for help or giving advice.

Fig. 6 The interface of the “defeat enemies” task



## Experiment design

### Participants

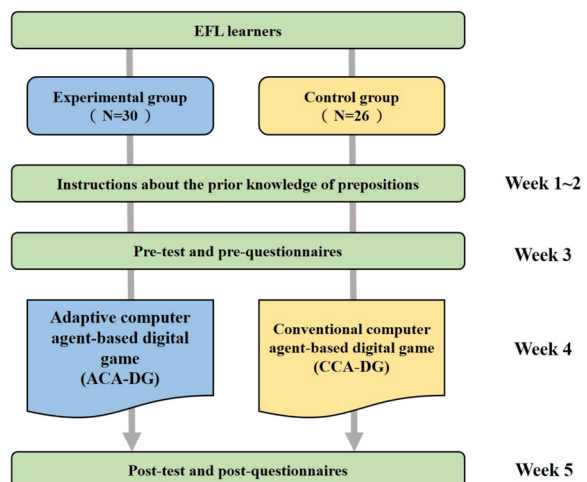
The current study aimed at improving English learning achievement through the adaptive computer agent-based digital game. We conducted a quasi-experimental design in a junior high school in northern Taiwan to investigate the effects of the game. The participants were 56 junior high school students from four classes. Their average age was 14. The ACA-DG group had 30 students and learned with the adaptive computer agent-based digital game; the CCA-DG group had 26 students and learned with the conventional computer agent-based digital game. All participants had the same level of computer skills training. They were taught by the same teacher who had taught English for over 6 years. These students participated in the experiments voluntarily and were informed that the results would not be assessed as part of their course grades.

### Experimental procedure

The experimental procedure is shown in Fig. 7. The whole experiment lasted for 5 weeks and in each week the class was 100 min. In the 1st and 2nd weeks, students were taught the prior knowledge of prepositions to ensure they could keep up with the instruction. The teacher also trained students to use basic computer skills to avoid obstacles during the game. Both groups of students were required to complete the pretest and pre-questionnaires on academic achievement, motivation, and self-efficacy in the 3rd week. Students entered the computer classroom in the 4th week to complete the digital game-based learning activities. In the last week, both groups of students were required to complete the posttest and post-questionnaires on academic achievement, motivation, self-efficacy, and English anxiety.

At the beginning of the game, the computer agent roles in the experimental group and the control group were respectively set to peer and teacher. The experimental group learned with the adaptive computer agent-based digital game (ACA-DG). When each task was completed, the system recorded and analyzed the players' game behaviors. If the decision

Fig. 7 Experimental procedure



tree model determined that the player had low learning achievement, the system would switch the computer agent role in the next task. The control group learned with the conventional computer agent-based digital game (CCA-DG). Regardless of the learner's behavior during the task, the computer agent maintained the teacher role throughout the game.

## Instruments

The instruments used in this study comprised tests and questionnaires on academic achievement, motivation, self-efficacy, and English anxiety. The pre- and posttest were designed by two experienced English teachers, and aimed at assessing EFL students' knowledge of time and place prepositions. The pretest consisted of 15 multiple-choice questions (4 points/question) and 10 fill-in-the-blank questions (4 points/question). The total score was 100 points. The KR20 value was 0.81. The post-test consisted of 10 multiple-choice questions (4 points/question), 10 cloze questions, and five fill-in-the-blank questions (4 points/question). The total score was 100 points. The KR20 value was 0.73.

The learning motivation questionnaire was modified from the measure developed by Wang and Chen (2010). The questionnaire used a 5-point Likert-type rating scheme (1 = *strongly disagree*; 5 = *strongly agree*) and consisted of six items. The overall Cronbach's alpha was 0.79. An example item of this measure is: "I prefer course material that really challenges me so I can learn new things."

The self-efficacy questionnaire was modified from the measure developed by Pintrich et al. (1991). The questionnaire used a 5-point Likert-type rating scheme (1 = *strongly disagree*; 5 = *strongly agree*) and consisted of eight items. The overall Cronbach's alpha was 0.93. An example item of this measure is: "I believe I will receive an excellent grade in this class."

The English anxiety questionnaire was modified from the foreign language classroom anxiety (FLCA) measure developed by Horwitz et al. (1986). The questionnaire used a 5-point Likert-type rating scheme (1 = *strongly disagree*; 5 = *strongly agree*) and consisted of 11 items. The overall Cronbach's alpha was 0.94. An example item of this measure is: "I don't worry about making mistakes in English class."

## Experimental results

### Learning achievement

This study employed the one-way analysis of covariance (ANCOVA) to evaluate the academic achievement of students from the two groups. The independent variable was the computer agent category; the dependent variable was the posttest score of learning achievement; the covariate was the pretest score of learning achievement. The Levene's test was not significant ( $F(1,54) = 3.80, p > 0.05$ ) and did not violate the homogeneity assumption. The result also passed the test of homogeneity of the regression coefficient ( $F(1,52) < 0.01, p > 0.05$ ), and therefore ANCOVA could be conducted. As shown in Table 3, the adjusted means for the experimental and control groups were 61.87 and 52.01. The two types of computer agents affected the students' academic achievement and produced a significant difference ( $F(1,53) = 4.15, p < 0.05$ ). The variation showed a medium effect ( $\eta^2 = 0.07$ ) on learning achievement. The results indicated that students

**Table 3** The one-way ANCOVA result of learning achievement

| Group              | <i>N</i> | <i>Mean</i> | <i>SD</i> | <i>Adjusted M</i> | <i>Adjusted SD</i> | <i>F</i> | $\eta^2$ |
|--------------------|----------|-------------|-----------|-------------------|--------------------|----------|----------|
| Experimental group | 30       | 59.47       | 19.02     | 61.87             | 3.25               | 4.15*    | 0.07     |
| Control group      | 26       | 54.77       | 21.81     | 52.01             | 3.39               |          |          |

\* $p < 0.05$ 

who studied with the ACA-DG had significantly higher academic achievement than those who studied with the CCA-DG. The adaptive computer agent-based digital game enhanced students' learning outcomes.

### Learning motivation

The one-way ANCOVA was employed to evaluate students' motivation. The independent variable was the computer agent category; the dependent variable was the posttest score of learning motivation; the covariate was the pretest score of learning motivation. The Levene's test was not significant ( $F(1,54) = 3.44, p > 0.05$ ) and did not violate the homogeneity assumption. The result also passed the test of homogeneity of the regression coefficient ( $F(1,52) = 1.95, p > 0.05$ ). As shown in Table 4, the adjusted means for the experimental and control groups were 3.56 and 3.82. There was no significant difference in the motivation of the two groups ( $F(1,53) = 1.71, p > 0.05$ ). The results indicated that students who studied with the ACA-DG had the same level of learning motivation as those who studied with the CCA-DG.

This study used the paired sample  $t$  test to further analyze the difference in learning motivation in the pre- and posttest. As shown in Table 5, the pretest mean was 3.34 and the posttest mean was 3.56 in the ACA-DG group. The result of the paired sample  $t$  test was significant ( $t(29) = -2.17, p < 0.05$ ). In the CCA-DG group, the pretest mean was 3.17 and the posttest mean was 3.82. The result of the paired sample  $t$  test was also significant ( $t(25) = -2.56, p < 0.05$ ) and indicated that both groups of students had increased motivation after using the computer agent-based digital games.

**Table 4** The one-way ANCOVA result of learning motivation

| Group              | <i>N</i> | <i>Mean</i> | <i>SD</i> | <i>Adjusted M</i> | <i>Adjusted SD</i> | <i>F</i> |
|--------------------|----------|-------------|-----------|-------------------|--------------------|----------|
| Experimental group | 30       | 3.56        | 0.47      | 3.56              | 0.14               | 1.71     |
| Control group      | 26       | 3.82        | 0.99      | 3.83              | 0.15               |          |

**Table 5** The paired sample  $t$ -test results of learning motivation

| Group              |           | <i>Mean</i> | <i>SD</i> | <i>t</i> | <i>Cohen's d</i> |
|--------------------|-----------|-------------|-----------|----------|------------------|
| Experimental group | Pre-test  | 3.34        | 0.52      | -2.17*   | 0.44             |
|                    | Post-test | 3.56        | 0.47      |          |                  |
| Control group      | Pre-test  | 3.17        | 0.74      | -2.56*   | 0.74             |
|                    | Post-test | 3.82        | 0.99      |          |                  |

\* $p < 0.05$

**Table 6** The one-way ANCOVA result of self-efficacy

| Group              | <i>N</i> | <i>Mean</i> | <i>SD</i> | <i>Adjusted M</i> | <i>Adjusted SD</i> | <i>F</i> | $\eta^2$ |
|--------------------|----------|-------------|-----------|-------------------|--------------------|----------|----------|
| Experimental group | 30       | 3.12        | 0.74      | 3.17              | 0.10               | 4.39*    | 0.08     |
| Control group      | 26       | 2.92        | 0.74      | 2.86              | 0.11               |          |          |

\* $p < 0.05$ **Table 7** The one-way ANCOVA result of English anxiety

| Group              | <i>N</i> | <i>Mean</i> | <i>SD</i> | <i>Adjusted M</i> | <i>Adjusted SD</i> | <i>F</i> | $\eta^2$ |
|--------------------|----------|-------------|-----------|-------------------|--------------------|----------|----------|
| Experimental group | 30       | 3.04        | 0.39      | 2.98              | 0.07               | 5.70*    | 0.10     |
| Control group      | 26       | 3.18        | 0.60      | 3.25              | 0.08               |          |          |

\* $p < 0.05$ 

## Self-efficacy

The one-way ANCOVA was employed to evaluate students' self-efficacy. The independent variable was the computer agent category; the dependent variable was the posttest score of learning motivation; the covariate was the pretest score. The Levene's test was not significant ( $F(1,54) = 2.29, p > 0.05$ ) and did not violate the homogeneity assumption. The result also passed the test of homogeneity of the regression coefficient ( $F(1,52) = 0.31, p > 0.05$ ). As shown in Table 6, the adjusted means for the experimental and control groups were 3.17 and 2.86. There was a significant difference in the self-efficacy of the two groups ( $F(1,53) = 4.39, p < 0.05$ ). The variation showed a medium effect ( $\eta^2 = 0.08$ ) on self-efficacy. The results indicated that students who studied with the ACA-DG had significantly higher self-efficacy than those who studied with the CCA-DG.

## English anxiety

In this study, one-way ANCOVA was used as the analysis method to evaluate two groups' English anxiety. The independent variable was the computer agent category; the dependent variable was the posttest score of English anxiety; the covariate was the pretest score. The results passed the Levene's test of determining homogeneity ( $F(1,54) = 2.34, p > 0.05$ ) and the test of homogeneity of the regression coefficient ( $F(1,52) < 0.01, p > 0.05$ ). As shown in Table 7, the adjusted means for the experimental and control groups were 2.98 and 3.25. There was a significant difference in the English anxiety of the two groups ( $F(1,53) = 5.70, p < 0.05$ ). The variation showed a medium-to-large effect ( $\eta^2 = 0.10$ ) on English anxiety. The results indicated that students who studied with the ACA-DG had significantly lower English anxiety than those who studied with the CCA-DG.

**Table 8** The distribution of computer agent role switching for learners with different cognitive levels in the ACA-DG group

| Cognitive level | Frequency of role switching |       |          |       |          |       | Total |
|-----------------|-----------------------------|-------|----------|-------|----------|-------|-------|
|                 | 0                           |       | 1        |       | 2        |       |       |
|                 | <i>N</i>                    | %     | <i>N</i> | %     | <i>N</i> | %     |       |
| High            | 3                           | 20.00 | 7        | 46.67 | 5        | 33.33 | 15    |
| Low             | 6                           | 40.00 | 6        | 40.00 | 3        | 20.00 | 15    |
| Total           | 9                           |       | 13       |       | 8        |       | 30    |

$$\chi^2 = 1.58, df = 2, p > 0.05$$

**Table 9** The direction of agent role switching for learners with different cognitive levels in the ACA-DG group

| Original role | Role after switching | Cognitive level |       |          |       | Total    |        |
|---------------|----------------------|-----------------|-------|----------|-------|----------|--------|
|               |                      | High            |       | Low      |       |          |        |
|               |                      | <i>N</i>        | %     | <i>N</i> | %     | <i>N</i> | %      |
| Teacher       | Teacher              | 2               | 3.33  | 3        | 5.00  | 5        | 8.33   |
|               | Peer                 | 1               | 1.67  | 2        | 3.33  | 3        | 5.00   |
|               | Student              | 1               | 1.67  | 1        | 1.67  | 2        | 3.33   |
| Peer          | Teacher              | 6               | 10.00 | 7        | 11.67 | 13       | 21.67  |
|               | Peer                 | 9               | 15.00 | 13       | 21.67 | 22       | 36.67  |
|               | Student              | 6               | 10.00 | 2        | 3.33  | 8        | 13.33  |
| Student       | Teacher              | 1               | 1.67  | 0        | 0     | 1        | 1.67   |
|               | Peer                 | 2               | 3.33  | 0        | 0     | 2        | 3.33   |
|               | Student              | 2               | 3.33  | 2        | 3.33  | 4        | 6.67   |
| Total         |                      | 30              | 50.00 | 30       | 50.00 | 60       | 100.00 |

## Learning behavior features

In this study, learners in the experimental group were divided into two groups of high achievement and low achievement based on their posttest scores. Each group had 15 students. The system recorded the frequency of computer agent role switching for the learners with different cognitive levels, as shown in Table 8. In the high achievement group, three students' computer agents did not change their roles, seven students' computer agents switched roles once, while five students' computer agents switched roles twice. In the low-achieving group, six students' computer agents did not change their roles, six students' computer agents switched roles once, and three students' computer agents switched roles twice. The chi-square results were insignificant ( $\chi^2 = 1.58, df = 2, p > 0.05$ ), indicating no significant difference in the frequency of switching among students with different cognitive levels. However, the descriptive statistics showed that more students in the high achievement group switched agent roles once (46.67%) or twice (33.33%), while more students in the low achievement group did not switch agent roles (40.00%) or switched only once (40.00%).

This study also compared the direction of agent role switching for learners with different cognitive levels in the experimental group, as shown in Table 9. Most of the agents in

the high and low achievement groups remained the same for peers (36.67%) or switched from peer to teacher (21.67%) and student (13.33%). However, in the high achievement group, a higher proportion of agents changed roles to teachers (1.67%) or remained the same as students (3.33%) than in the low achievement group. The ratio of computer agents in the low achievement group who kept peer roles unchanged (21.67%) was higher than that in the high achievement group (15.00%).

In this study, the independent sample *t* test was used to compare the behavioral features of learners in the experimental and control groups, as shown in Table 10. Regarding cognitive status, the mean number of incorrect answers was 25.67 for the ACA-DG group and 32.38 for the CCA-DG group. The independent sample *t* test was significant ( $t = -2.17, p < 0.05$ ), indicating that the ACA-DG group had significantly fewer incorrect answers than the CCA-DG group. Regarding help-seeking behavior, the mean times for the ACA-DG and CCA-DG groups were 16.80 and 13.84. The independent sample *t* test was significant ( $t = 2.45, p < 0.05$ ). The results indicated that the ACA-DG group sought help significantly more often than the CCA-DG group. For the number of non-task interactions, the mean was 60.83 for the ACA-DG group and 53.54 for the CCA-DG group. The independent sample *t* test was significant ( $t = 2.30, p < 0.05$ ), indicating that learners in the ACA-DG group had significantly more non-task interactions than those in the CCA-DG group. Finally, regarding the knowledge review, the mean time was 1.80 for the ACA-DG group and 1.69 for the CCA-DG group. The independent sample *t* test was not significant ( $t = 0.34, p > 0.05$ ). There was therefore no significant difference in the knowledge review of the two groups.

## Discussion and conclusions

This was a quasi-experimental study to investigate the effectiveness of the adaptive computer agent-based digital game on EFL students' learning. The experimental group learned with the adaptive computer agent-based digital game (ACA-DG). The control group learned with the conventional computer agent-based digital game (CCA-DG). The results showed that students in the ACA-DG group had significantly higher learning achievement and self-efficacy than those in the CCA-DG group. On the other hand, students in the ACA-DG group had significantly lower English anxiety than those in the CCA-DG group. There was no significant difference between the two groups regarding learning motivation.

**Table 10** The *t*-test results of the behavior features

| Behavior feature     | Group  | <i>N</i> | <i>Mean</i> | <i>SD</i> | <i>t</i> | <i>Cohen's d</i> |
|----------------------|--------|----------|-------------|-----------|----------|------------------|
| Cognitive status     | ACA-DG | 30       | 25.67       | 8.92      | -2.51*   | 0.67             |
|                      | CCA-DG | 26       | 32.38       | 11.12     |          |                  |
| Help-seeking         | ACA-DG | 30       | 16.80       | 4.53      | 2.45*    | 0.66             |
|                      | CCA-DG | 26       | 13.84       | 4.45      |          |                  |
| Non-task interaction | ACA-DG | 30       | 60.83       | 11.00     | 2.30*    | 0.61             |
|                      | CCA-DG | 26       | 53.54       | 12.73     |          |                  |
| Knowledge review     | ACA-DG | 30       | 1.80        | 1.19      | 0.34     |                  |
|                      | CCA-DG | 26       | 1.69        | 1.16      |          |                  |

\* $p < 0.05$

In terms of English language learning achievement, learners who used the ACA-DG had significantly higher achievement than those who used the CCA-DG. The results of this study are consistent with the findings of other scholars. Providing content-related scaffolding based on learners' abilities helps them self-regulate and improve their learning outcomes (Rahimi et al., 2021). In the ACA-DG condition, learners with lower knowledge acquisition were guided to computer agents with teacher or peer roles. They had more opportunities to receive support and review suggestions about the learning content. Learners with better knowledge acquisition were directed to the peer agent or student agent model and had more opportunities for social interaction with the agents. In addition, the reflection and self-regulation in the peer and student agent models played significant roles in knowledge retention. In the CCA-DG condition, both types of learners interacted with the teacher computer agent. Learners with lower knowledge acquisition received immediate cues about the correct answer when they encountered a dilemma. However, this may have led to a reliance on external support and a lack of memory and internalization of knowledge. Learners with worse cognitive status need more problem-solving solutions or review suggestions to avoid making mistakes repeatedly (ter Vrugte et al., 2017). Thus, learners in the ACA-DG group had better learning achievement than those in the CCA-DG group. The present results further support the suggestion of several scholars that student-centered and adaptive learning models account for individual differences and improve the learning experience (Liu & Graf, 2009; Schrader & Grassinger, 2021).

There was no significant difference in the learning motivation of the experimental and control groups. Both the adaptive and conventional computer agents provided positive feedback, such as encouragement or understanding. The positive feedback and suggestions increased learners' confidence when they encountered difficulties. Therefore, there was no significant difference in the learning motivation of the two groups of learners in the posttest. Nevertheless, the paired sample *t* test showed that the learning motivation scores of both groups in the posttest were significantly higher than in the pretest. It is consistent with Reinders and Wattana's (2015) finding that rich multimedia materials and interaction in digital games promote students' learning engagement.

In terms of self-efficacy, the students who used ACA-DG had significantly higher self-efficacy than those who used CCA-DG. According to the social cognitive theory proposed by Bandura (2012), learners' perceptions in a game-based learning environment are divided into two types: learning and playing. Students with stronger learning perceptions have higher engagement and self-efficacy than those with gaming perceptions (Lu & Lien, 2020). Adaptive computer agents adjust how they intervene in the learning process based on the learner's characteristics. They help students with a strong gaming perception but a weak learning perception to focus on instructional goals to enhance their engagement. Bilgin et al. (2015) suggested that increasing opportunities to express concepts and share experiences in a collaborative context improves students' self-efficacy. Positive feedback gives students more confidence to complete learning tasks (Chen et al., 2020). The conventional computer agents work in a single, instructional, and supervisory way that does not fully function for social interaction with the learners. Sawyer et al. (2017) also suggested that lower freedom in game-based learning may have a negative impact on students' engagement and self-efficacy in games.

The study also compared English anxiety after completing the learning activities. The results revealed that students in the ACA-DG group had significantly lower English anxiety than students in the CCA-DG group. Adaptive computer agents provide various kinds of positive feedback to build emotional bonds with students when they fail to answer questions or complete tasks. For example, peer agents show more tolerance in

games. Student agents help build learners' confidence by expressing admiration. Johnson and Lester (2018) suggested that agents who show empathy deepen their emotional bonds with learners and reduce their negative emotions. Franciosi (2017) believes that digital games with more freedom and encouragement offer learners a more relaxed linguistic environment. The interactive games strengthen the learners' communicative intentions and reduce their language anxiety (Reinder & Wattana, 2015). On the other hand, the computer agents in the CCA-DG group appeared in the teacher role. When students encountered difficulties, they received a single solution with a high cognitive load. These findings reinforce those of previous studies. The current results support those of previous studies. For example, a non-pressured digital game environment reduces the learning anxiety of learners with lower cognitive levels (Calvo-Ferrer, 2017; Wei et al., 2018).

In terms of behavioral analysis, high achieving learners in the ACA-DG group had more opportunities to switch roles than low achieving students. When learners' cognitive status improved, the system tended to change the agent role to peer and student. Previous research has shown that a bidirectional reciprocal learning model boosts students' confidence in completing tasks (Chen et al., 2020). At the same time, agents who showed empathy also reduced students' learning anxiety (Arguedas & Daradoumis, 2021). The chi-squared results showed no significant difference between high- and low-achieving students in terms of the number of role switches. The smaller sample size may have influenced this result.

Finally, in terms of behavior features, learners in the ACA-DG group had a significantly better correct question rate than those in the CCA-DG group. This result was consistent with the posttest scores of learning achievement. Learners with better cognitive status in the game also scored higher on the posttest. The CCA-DG group sought help significantly more often than the ACA-DG group. This result supports Liu and Graf's (2009) finding that adaptive learning approaches provide guidance and cues to students who need help. In addition, the number of non-task interactions was significantly higher in the experimental group than in the control group, indicating that ACA-DG facilitates students' social interactions with the environment or virtual characters in the game. Diverse emotional feedback increases students' curiosity, creativity, and perception of learning texts in virtual environments (Chen et al., 2020; Johnson & Lester, 2018).

Although this study supports the effectiveness of ACA-DG, there are some research limitations. First, this study was affected by the coronavirus epidemic, and the sample size was not large enough. Therefore, whether the results of the behavioral analysis can be extrapolated to other courses needs to be further discussed. Secondly, the role-switching strategy developed in this study was based on the expert decision. Third, the experiment time was only one week, and hence one needs to be cautious when making inferences based on the findings. In the future, we will try to collect more learners' behavioral data for machine learning and add monitoring of learners' feelings to form a more accurate agent role-switching decision tree. The interaction and behavior patterns of the teacher, peer, and student computer agents used in this study are also worth further adjustment and supplementation. Moreover, we hope to conduct longer experiments in the future to further investigate the effectiveness of the proposed approach in various dimensions.

In summary, according to the experimental results, the adaptive computer agent-based digital game developed in this study helped learners gain better learning achievement, higher self-efficacy, and lower English learning anxiety. The adaptive computer agent mechanism positively affected the learners who used digital games. It is hoped that the results of this study will provide more references for future digital game development.

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**Data availability** The data of this study can be provided upon sending requests to the corresponding author.

## Declarations

**Conflict of interest** The authors would like to declare that there is no conflict of interest in this study.

**Ethical approval** The participants were protected by hiding their personal information during the research process. They knew that the participation was voluntary and they could withdraw from the study at any time.

## References

- Arguedas, M., & Daradoumis, T. (2021). Analysing the role of a pedagogical agent in psychological and cognitive preparatory activities. *Journal of Computer Assisted Learning*, 37(4), 1167–1180.
- Atkinson, R. K., Mayer, R. E., & Merrill, M. M. (2005). Fostering social agency in multimedia learning: Examining the impact of an animated agent’s voice. *Contemporary Educational Psychology*, 30(1), 117–139.
- Bandura, A. (2012). On the functional properties of perceived self-efficacy revisited. *Journal of Management*, 38(1), 9–44.
- Belpaeme, T., Kennedy, J., Ramachandran, A., Scassellati, B., & Tanaka, F. (2018). Social robots for education: A review. *Science Robotics*, 3(21), eaat5954.
- Bevacqua, E., Eyben, F., Heylen, D., ter Maat, M., Pammi, S., Pelachaud, C., & Wöllmer, M. (2011). Interacting with emotional virtual agents. *International Conference on Intelligent Technologies for Interactive Entertainment*. Springer.
- Bilgin, C. U., Baek, Y., & Park, H. (2015). How debriefing strategies can improve student motivation and self-efficacy in game-based learning. *Journal of Educational Computing Research*, 53(2), 155–182.
- Bunyakul, N., Wiwatwattana, N., & Panjaburee, P. (2022). Effects of a mobile game on students’ learning achievements and motivations in a clinical chemistry course: Learning style differences. *International Journal of Mobile Learning and Organisation*, 16(2), 221–244.
- Cai, Z., Feng, S., Baer, W., & Graesser, A. (2014). Instructional strategies in Trialogue-based intelligent tutoring systems. *Design Recommendations for Intelligent Tutoring Systems*, 2, 225–235.
- Calvo-Ferrer, J. R. (2017). Educational games as stand-alone learning tools and their motivational effect on L 2 vocabulary acquisition and perceived learning gains. *British Journal of Educational Technology*, 48(2), 264–278.
- Castro-Alonso, J. C., Wong, R. M., Adesope, O. O., & Paas, F. (2021). Effectiveness of multimedia pedagogical agents predicted by diverse theories: A meta-analysis. *Educational Psychology Review*, 33(3), 989–1015.
- Chen, H. J. H., & Hsu, H. L. (2020). The impact of a serious game on vocabulary and content learning. *Computer Assisted Language Learning*, 33(7), 811–832.
- Chen, H., Park, H. W., & Breazeal, C. (2020). Teaching and learning with children: Impact of reciprocal peer learning with a social robot on children’s learning and emotive engagement. *Computers & Education*, 150, 103836.
- Daunghcharone, K., Panjaburee, P., & Thongkoo, K. (2020). Implementation of mobile game-transformed lecture-based approach to promoting C programming language learning. *International Journal of Mobile Learning and Organisation*, 14(2), 236–254.
- Davis, R. O., Vincent, J., & Wan, L. (2021). Does a pedagogical agent’s gesture frequency assist advanced foreign language users with learning declarative knowledge? *International Journal of Educational Technology in Higher Education*, 18(1), 1–19.
- Dinçer, S., & Doğanay, A. (2017). The effects of multiple-pedagogical agents on learners’ academic success, motivation, and cognitive load. *Computers & Education*, 111, 74–100.

- El-Sabagh, H. A. (2021). Adaptive e-learning environment based on learning styles and its impact on development students' engagement. *International Journal of Educational Technology in Higher Education*, 18(1), 1–24.
- Franciosi, S. J. (2017). The effect of computer game-based learning on FL vocabulary transferability. *Educational Technology & Society*, 20(1), 123–133.
- Graesser, A. C., Forsyth, C. M., & Lehman, B. A. (2017). Two heads may be better than one: Learning from computer agents in conversational dialogues. *Grantee Submission*, 119, 1–20.
- Hanus, M. D., & Fox, J. (2015). Assessing the effects of gamification in the classroom: A longitudinal study on intrinsic motivation, social comparison, satisfaction, effort, and academic performance. *Computers & Education*, 80, 152–161.
- Horwitz, E. K., Horwitz, M. B., & Cope, J. (1986). Foreign language classroom anxiety. *The Modern Language Journal*, 70(2), 125–132.
- Howard, C., Jordan, P., Di Eugenio, B., & Katz, S. (2017). Shifting the load: A peer dialogue agent that encourages its human collaborator to contribute more to problem solving. *International Journal of Artificial Intelligence in Education*, 27(1), 101–129.
- Hung, H. T., Yang, J. C., Hwang, G. J., Chu, H. C., & Wang, C. C. (2018). A scoping review of research on digital game-based language learning. *Computers & Education*, 126, 89–104.
- Hwang, G. J., Sung, H. Y., Chang, S. C., & Huang, X. C. (2020). A fuzzy expert system-based adaptive learning approach to improving students' learning performances by considering affective and cognitive factors. *Computers and Education: Artificial Intelligence*, 1, 100003.
- Israel-Fishelson, R., & Hershkovitz, A. (2021). Micro-persistence and difficulty in a game-based learning environment for computational thinking acquisition. *Journal of Computer Assisted Learning*, 37(3), 839–850.
- Jiang, L. C., Bazarova, N. N., & Hancock, J. T. (2011). The disclosure–intimacy link in computer-mediated communication: An attributional extension of the hyperpersonal model. *Human Communication Research*, 37(1), 58–77.
- Johnson, W. L., & Lester, J. C. (2018). Pedagogical agents: Back to the future. *AI Magazine*, 39(2), 33–44.
- Karampiperis, P., & Sampson, D. (2005). Adaptive learning resources sequencing in educational hypermedia systems. *Educational Technology & Society*, 8(4), 128–147.
- Kim, B., Park, H., & Baek, Y. (2009). Not just fun, but serious strategies: Using meta-cognitive strategies in game-based learning. *Computers & Education*, 52(4), 800–810.
- Komalawardhana, N., Panjaburee, P., & Srisawasdi, N. (2021). A mobile game-based learning system with personalised conceptual level and mastery learning approach to promoting students' learning perceptions and achievements. *International Journal of Mobile Learning and Organisation*, 15(1), 29–49.
- Liu, T. C., & Graf, S. (2009). Coping with mismatched courses: Students' behaviour and performance in courses mismatched to their learning styles. *Educational Technology Research and Development*, 57(6), 739.
- Lu, Y. L., & Lien, C. J. (2020). Are they learning or playing? Students' perception traits and their learning self-efficacy in a game-based learning environment. *Journal of Educational Computing Research*, 57(8), 1879–1909.
- Ma, W., Adesope, O. O., Nesbit, J. C., & Liu, Q. (2014). Intelligent tutoring systems and learning outcomes: A meta-analysis. *Journal of Educational Psychology*, 106(4), 901.
- Martha, A. S. D., & Santoso, H. B. (2019). The design and impact of the pedagogical agent: A systematic literature review. *Journal of Educators Online*, 16(1), n1.
- Maslow, A. H. (1943). A theory of human motivation. *Psychological Review*, 50(4), 370–396.
- Mostefai, B., Balla, A., & Trigano, P. (2019). A generic and efficient emotion-driven approach toward personalized assessment and adaptation in serious games. *Cognitive Systems Research*, 56, 82–106.
- Okita, S. Y., & Schwartz, D. L. (2013). Learning by teaching human pupils and teachable agents: The importance of recursive feedback. *Journal of the Learning Sciences*, 22(3), 375–412.
- Pacella, D., & López-Pérez, B. (2018). Assessing children's interpersonal emotion regulation with virtual agents: The serious game Emodiscovery. *Computers & Education*, 123, 1–12.
- Pai, K. C., Kuo, B. C., Liao, C. H., & Liu, Y. M. (2021). An application of Chinese dialogue-based intelligent tutoring system in remedial instruction for mathematics learning. *Educational Psychology*, 41(2), 137–152.
- Pintrich, P. R., Smith, D. A. F., Garcia, T., & McKeachie, W. J. (1991). *A manual for the use of the motivated strategies for learning questionnaire (MSLQ)*. MI: National Center for Research to Improve Postsecondary Teaching and Learning. (ERIC Document Reproduction Service No. ED 338122)
- Qian, M., & Clark, K. R. (2016). Game-based Learning and 21st century skills: A review of recent research. *Computers in Human Behavior*, 63, 50–58.

- Rahimi, S., Shute, V., Kuba, R., Dai, C. P., Yang, X., Smith, G., & Fernández, C. A. (2021). The use and effects of incentive systems on learning and performance in educational games. *Computers & Education*, 165, 104135.
- Reinders, H., & Wattana, S. (2015). Affect and willingness to communicate in digital game-based learning. *ReCALL*, 27(1), 38–57.
- Reinhardt, J. (2017). Digital gaming in L2 teaching and learning. In C. A. Chapelle & S. Sauro (Eds.), *The handbook of technology and second language teaching and learning*. Wiley.
- Richards, D., & Dignum, V. (2019). Supporting and challenging learners through pedagogical agents: Addressing ethical issues through designing for values. *British Journal of Educational Technology*, 50(6), 2885–2901.
- Sawyer, R., Smith, A., Rowe, J., Azevedo, R., & Lester, J. (2017). Is more agency better? The impact of student agency on game-based learning. *International Conference on Artificial Intelligence in Education* (pp. 335–346). Springer.
- Schrader, C., & Grassinger, R. (2021). Tell me that I can do it better. The effect of attributional feedback from a learning technology on achievement emotions and performance and the moderating role of individual adaptive reactions to errors. *Computers & Education*, 161, 104028.
- Schroeder, N. L., & Gotch, C. M. (2015). Persisting issues in pedagogical agent research. *Journal of Educational Computing Research*, 53(2), 183–204.
- Schroeder, N. L., Romine, W. L., & Craig, S. D. (2017). Measuring pedagogical agent persona and the influence of agent persona on learning. *Computers & Education*, 109, 176–186.
- Shute, V., Rahimi, S., Smith, G., Ke, F., Almond, R., Dai, C. P., & Sun, C. (2021). Maximizing learning without sacrificing the fun: Stealth assessment, adaptivity and learning supports in educational games. *Journal of Computer Assisted Learning*, 37(1), 127–141.
- Standen, P. J., Brown, D. J., Taheri, M., Galvez Trigo, M. J., Boulton, H., Burton, A., & Hortal, E. (2020). An evaluation of an adaptive learning system based on multimodal affect recognition for learners with intellectual disabilities. *British Journal of Educational Technology*, 51(5), 1748–1765.
- Sun, C. T., Chen, L. X., & Chu, H. M. (2018). Associations among scaffold presentation, reward mechanisms and problem-solving behaviors in game play. *Computers & Education*, 119, 95–111.
- Tanaka, M. (2017). Factors affecting motivation for short in-class extensive reading. *Journal of Asia TEFL*, 14(1), 98.
- Taub, M., Sawyer, R., Smith, A., Rowe, J., Azevedo, R., & Lester, J. (2020). The agency effect: The impact of student agency on learning, emotions, and problem-solving behaviors in a game-based learning environment. *Computers & Education*, 147, 103781.
- ter Vrugte, J., de Jong, T., Vandercruysse, S., Wouters, P., van Oostendorp, H., & Elen, J. (2017). Computer game-based mathematics education: Embedded faded worked examples facilitate knowledge acquisition. *Learning and Instruction*, 50, 44–53.
- Tsai, C. H., Cheng, C. H., Yeh, D. Y., & Lin, S. Y. (2017). Can learning motivation predict learning achievement? A case study of a mobile game-based English learning approach. *Education and Information Technologies*, 22(5), 2159–2173.
- Wang, L. C., & Chen, M. P. (2010). The effects of game strategy and preference-matching on flow experience and programming performance in game-based learning. *Innovations in Education and Teaching International*, 47(1), 39–52.
- Wang, M., & Zheng, X. (2021). Using game-based learning to support learning science: A study with middle school students. *The Asia-Pacific Education Researcher*, 30(2), 167–176.
- Wei, C. W., Kao, H. Y., Lu, H. H., & Liu, Y. C. (2018). The effects of competitive gaming scenarios and personalized assistance strategies on English vocabulary learning. *Educational Technology & Society*, 21(3), 146–158.
- Yadav, A. K., & Oyelere, S. S. (2021). Contextualized mobile game-based learning application for computing education. *Education and Information Technologies*, 26(3), 2539–2562.
- Yang, J. C., Lin, M. Y. D., & Chen, S. Y. (2018). Effects of anxiety levels on learning performance and gaming performance in digital game-based learning. *Journal of Computer Assisted Learning*, 34(3), 324–334.
- Yang, J. C., Quadir, B., & Chen, N. S. (2016). Effects of the badge mechanism on self-efficacy and learning performance in a game-based English learning environment. *Journal of Educational Computing Research*, 54(3), 371–394.
- Zeng, H., Zhou, S. N., Hong, G. R., Li, Q. Y., & Xu, S. Q. (2020). Evaluation of interactive game-based learning in physics domain. *Journal of Baltic Science Education*, 19(3), 484–498.
- Zou, D., Huang, Y., & Xie, H. (2021). Digital game-based vocabulary learning: Where are we and where are we going? *Computer Assisted Language Learning*, 34(5–6), 751–777.

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